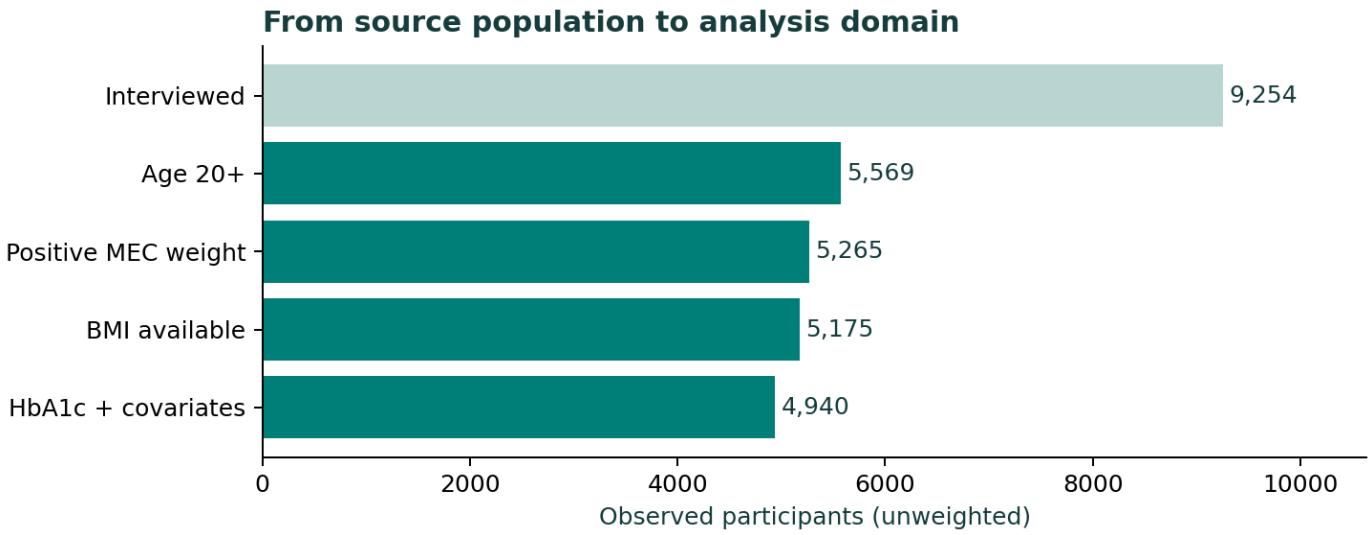
BMI and HbA1c in U.S. adults

NHANES 2017-2018 | Adults aged 20+ | Four official CDC components

+0.125 pp

Adjusted HbA1c difference per 5 kg/m² higher BMI. 95% CI 0.103 to 0.146; n=4,940.

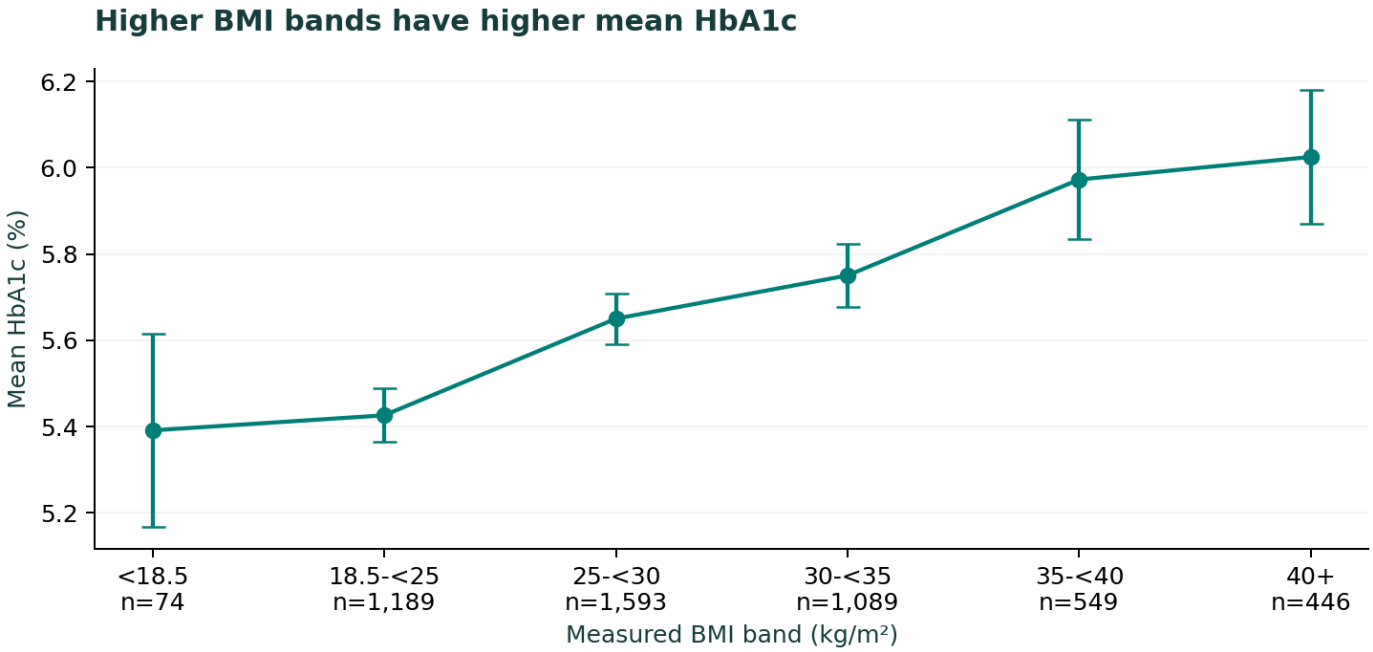
A positive association persists after adjustment for age and recorded sex. This estimate does not show that changing BMI causes a change in HbA1c.



Question and methods. DEMO_J is the participant spine; GHB_J supplies HbA1c, BMX_J supplies BMI, and DIQ_J supplies reported diabetes status. Keys and joins are validated before analysis. The full positive-weight MEC design is built before adult and complete-case domains.

Inference. R survey uses WTMEC2YR, SDMVSTRA and nested SDMVPSU, with Taylor linearization. Means use domain design t intervals; regression uses residual design degrees of freedom. Weighting does not correct all complete-case selection.

Results, checks and limits



Points are survey-weighted complete-case domain means; bars are 95% t intervals. Unweighted sample sizes are printed below each BMI band. These are descriptive means, not adjusted predictions.

Sensitivity. In the prespecified DIQ010=2 domain (no reported diabetes), the BMI coefficient is 0.075 pp (95% CI 0.061-0.090; n=3,994). This domain excludes borderline and unknown answers and is not all adults.

Verification. Python normal equations reproduce the primary coefficients; an independent stratum/PSU Taylor calculation reproduces both mean standard errors. All eight reconciliation checks pass. Corrupt keys, schema failures, invalid categories and domain calculations have executable tests.

Limitations. Cross-sectional association; no temporal or causal inference. Complete-case selection may bias estimates. MEC weights address the sample design, not all item missingness. Age is top-coded at 80; linear age adjustment may leave residual confounding. Treatment, lifestyle and other unmeasured factors are not adjusted. Sparse BMI bands have wider intervals. Results describe the 2017-2018 cycle, not current health status. The analysis is an independent public-data portfolio project, unaffiliated with Kaiser Permanente; no clinical recommendations.

Sources. CDC/NCHS NHANES 2017-2018 DEMO_J, GHB_J, BMX_J, DIQ_J. Exact URLs, SHA256 hashes, codebooks, run logs and aggregate tables are included in the repository. Data use is limited to statistical analysis and reporting.